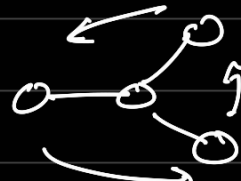


* SO(9): SO(11-2)

V_μ	$[1000]_9$	<u>9</u>	$D = 11$ massless irreps
$A^{(2)}$	$[0100]_9$	<u>36</u>	
$A^{(3)}$	$[0010]_9$	<u>84</u>	
χ_α	$[0001]_9$	<u>16</u>	Majorana
$h_{\mu\nu}$	$[2000]_9$	<u>44</u>	
ψ_α	$[1001]_9$	<u>128</u>	D_4 ?

SO(8):

V_μ	$[1000]_8$	<u>8</u>	triviality
$A^{(2)}$	$[0100]_8$	<u>28</u>	
χ_α	$[0010]_8$	<u>8</u>	
$\psi^{\dot{\alpha}}$	$[0001]_8$	<u>8</u>	



- Invariant under (n_1, n_2, n_3) permutations
- Before being bagged down by the field theory content lets look at char rep of the rep theory:

$$SO(8) \subset SO(9)$$

→ Every rep of $SO(9)$ projects into a sum of irreps of $SO(8)$

Branching rules: R : rep of $SO(9)$
 r_i : rep of $SO(8)$

$$R = \sum_i r_i$$

ex:

$$\begin{aligned} [1000]_9 &= [1000]_8 \oplus [0000]_8 & \square \\ V_\mu & \quad V_\mu & \quad V_9 \\ \underline{9} & \quad \underline{8} & \quad \underline{1} \end{aligned}$$

$$\begin{aligned} [0100]_9 &= [0100]_8 \oplus [1000]_8 & \square \cong \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \oplus \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ A_{\mu\nu} & \quad A_{\mu\nu} & \oplus & A_{9\mu} \end{aligned}$$

• Are not stable multiplets

Def: The p^{th} symmetric product of an IRREP $[n_1 \dots n_r]_n$ is denoted $\text{Sym}^p [n_1 \dots n_r]_n$
 The p^{th} antisymm. product $\Lambda^p [n_1 \dots n_r]_n$

$$\text{Sym}^p (R_1 \oplus R_2) = \bigoplus_{k=0}^p (\text{Sym}^k (R_1) \otimes \text{Sym}^{p-k} (R_2))$$

$$\Lambda^p (R_1 \oplus R_2) = \bigoplus_{k=0}^p \Lambda^k (R_1) \otimes \Lambda^{p-k} (R_2)$$

* Note:

$$\Lambda^k [10 \dots 0] = \Lambda^{n-k} [10 \dots 0]$$

for $SO(n)$, so k can go up to $k_{\text{max}} = \frac{n}{2}$

$$\Lambda^k [10 \dots 0] \rightarrow A^{(k)}$$

$$\text{then } F^{k+1} = dA^{(k)}$$

$$\bullet \text{ In } \dim(\mathcal{H}) = n, \quad *F^{k+1} = G^{n-k-1} = dC^{n-k-2}$$

Hmm... not quite

$$\text{For } SO(2m) \quad \Lambda^{n \leq m-2} [10 \dots 0]_{2m} = [0 \dots 1 \dots 0]_{2m} \quad n_k = 2$$

$$\Lambda^{m-1} [10 \dots 0]_{2m} = [0 \dots 1 \dots 1]_{2m}$$

inequivalent spinor reps

$$\Lambda^m [10 \dots 0]_{2m} = [0 \dots 2]_{2m} \oplus [0 \dots 20]_{2m}$$

F
 $\underbrace{\hspace{1cm}}$
 Dual

$*F$
 $\underbrace{\hspace{1cm}}$
 anti-self dual

$$*F \wedge F$$

$$\bullet SO(2m+1), \text{ rank } m$$

$$\Lambda^{n \leq m-1} [10 \dots 0]_{2m+1} = [0 \dots 1 \dots 0]_{2m+1}$$

$$n_k = 1$$

$$\Lambda^{n+m} [10 \dots 0]_{2m+1} = [0 \dots 2]_{2m+1}$$

~ spinor rep

$$\text{Sym}^2 [1000]_9 = \underbrace{[2000]_9}_{\text{Symmetric traceless}} + \underbrace{[0000]_9}_{\text{trace}}$$

$$\Lambda^2 [1000] = [0100]_9$$

examples: Branching Rules of $[1000]_9$

• These rules allow us to understand the field content of e_n^a

$$\begin{aligned} \text{Sym}_2^2 [1000]_9 &= \text{Sym}^2 ([1000]_8 \oplus [0000]_8) \\ &= \bigoplus_{k=0}^2 \text{Sym}^k ([1000]_8) \oplus \text{Sym}^{2-k} ([0000]_8) \end{aligned}$$

$$\begin{aligned} &= \text{Sym}^1 [1000]_8 \oplus \text{Sym}^1 [0000]_8 + \text{Sym}^2 ([1000]_8) \oplus [0000] \\ &= [1000]_8 \oplus [0000]_8 + ([2000]_8 + [0000]_8) \oplus [0000] \\ &= ([1000]_8 \oplus [2000]_8 \oplus [0000]_8) \oplus [0000]_8 \end{aligned}$$

But

$$\text{Sym}^2 [1000]_9 = [2000]_9 \oplus [0000]_9$$

$$\therefore [2000]_9 = [2000]_8 \oplus [1000]_8 \oplus [0000]_8$$

• Similarly $\Lambda^2 [1000]_9 = [0100]_9$

$$\begin{aligned} &= \Lambda^2 [1000]_9 = \Lambda^2 [1000]_8 \Lambda^2 [0000]_8 \oplus \Lambda^2 [1000]_8 \Lambda^1 [000]_8 \\ &= ([1000]_8 \oplus [0200]_8) [0000]_8 \end{aligned}$$

∴

$$[0100]_9 = ([1000]_8 \oplus [0200]_8) [0000]_8$$

• Do not get explicit scalar since $\Lambda_{00} = 0$

• $[0010]_9 = \Lambda^3 [1000]_9$

$$\Lambda^3 [1000]_9 = \Lambda^3 ([1000]_8 \oplus [0000]_8)$$

$$= \Lambda^2[1000] \oplus \Lambda^2[0000] \oplus \Lambda^2[1000]_8 \oplus \Lambda^3[1000]$$

$$= [0100]_8 \oplus \Lambda^3[1000]_8$$

$$\left[\begin{array}{l} \psi_{\alpha i} \psi_{\alpha} \sim \sigma_{\alpha i}^n V_M \oplus \epsilon_{\alpha \dot{\beta}} (\gamma_{\mu\nu\sigma})^{\dot{\beta}\beta} \epsilon_{\alpha\beta} (\chi\psi) \\ [0000][0001] \sim [1000] \oplus [0011] \\ \quad \quad \quad \underline{8} \qquad C_{\mu\nu\sigma} \end{array} \right]$$

$$\therefore \Lambda^3 [100]_g = [0010] = [0100]_g + [0011]_g$$

Sprünge: χ_α $[0001]_g = [0010]_8 + [0001]_8$

- Splits into irreps of of L & R handed spinors of $SO(8)$.

$$[1000]_q \oplus [0001]_q = [1001]_q \oplus [0001]_q$$

$$V \otimes \chi = \psi_\alpha^m \oplus \gamma_\mu^m \chi_\alpha ?$$

$$\Rightarrow [1001]_9 + 0002]_9$$

$$= ([1000]_8 \oplus [0000]_8) ([10010]_8 + [0001]_8)$$

$$= [1000]_8 [0010]_8^2 + [0000] [0010]_8 + [1000]_8 [0001]_8 + [0008]_8 [0001]_8$$

which we break down

$$[100]_2 [100]_2 = [2000]_2 + [0100]_2 + [0000]_2$$

$$[0010]_8 [0001]_8 = [0011]_8 + [1000]_8$$

$$[100]_8 [0010]_8 = [1010]_8 + [0001]_8$$

$$[1000]_8 [0001]_8 = [1001]_8 + [0010]_8$$

* Used trial/ty to find the products
thus

$$[1001]_9 = [1010]_8 + [1001]_8 + [0001]_8 + [0010]_8$$

$$\begin{aligned} 11D \quad & \psi_\mu^a [100]_9 \\ & A^{(3)} [0010]_9 \\ & e_\mu^a [2000]_9 \end{aligned}$$

(11D Supergrav. Multiplet)

$$\begin{aligned} & ([100]_9 \oplus [0010]_9 \oplus [2000]_9) \\ & = ([1010]_8 + [1001]_8 + [0010]_8 + [0001]_8) \\ & + ([0100]_8 + [0011]_8) \\ & + ([2000]_8 + [1000]_8 + [0000]_8) \end{aligned}$$

* From dimensional reduction of 11-D super we obtain the D=10 Type IIA supermultiplet w/ 32 supercharges.
 $N = (1, 2)$ susy

$$G_{IIA} = \begin{array}{cccccccccc} [2000]_8 & [1000]_8 & [0000]_8 & [0011]_8 & [0100]_8 & [1010]_8 & [1001]_8 & [0001]_8 & [0010]_8 & \\ \text{graviton} & \text{1-form} & \text{Dilaton} & \text{3-form} & \text{NS 2-form} & \text{LH grav.} & \text{RH grav.} & \text{RH spinor} & \text{LH spinor} & \\ e_\mu^m & B_\mu & \phi & A^{(3)} & A^{(2)} & \psi_{\mu a} & \psi_\mu^{\dot{a}} & \lambda_{\dot{a}} & \chi^a & \end{array}$$

* Can be written as the product of two opposite chirality supermultiplets, each with 16 supercharges

$$\begin{aligned} V_{10}^L &= [1000]_8 + [0010]_8 \\ V_{10}^R &= [1000]_8 + [0001]_8 \\ G_{IIA} &= V_{10}^L V_{10}^R \end{aligned} \quad \left(\begin{array}{l} \text{Vector multiplets} \\ \text{Con-shell} \end{array} \right)$$

$$* \quad G_{IIA} = ([1000]_8 + [0010]_8) ([1000]_8 + [0001]_8) \quad *$$

(1, 2)

• phenomenological interpretation:

An observer on the worldsheet of the closed string sees decoupled left and right moving modes.

* We now see what our field content is from dim reg of the 10D theory. Will return to this when we look at branes.

Type IIA: Has maximal susy. Closed string theory.

• dim of Clifford alg. $2^{10/2} = \underline{32}$

$$B^T = B, \quad C = -C^T, \quad E = 1$$

10d: can also have Mag. - weyl

• M-theory

$$\downarrow \delta 1$$

(Type IIA) low energy effective action/theories corresponding to type II string theory

$$\left[(\text{Type I} \oplus \mathcal{N}=1 \text{ SYM}) \rightarrow (E_8 \oplus E_8 \text{ or } SO(32)) \right] \\ \text{Heterotic string}$$

Type IIA: Q_α , $\alpha = 1, \dots, 32$

$$\{Q_\alpha, Q_\beta\} = (\gamma^M C^{-1})_{\alpha\beta} P_M + (\gamma^{MN} C^{-1})_{\alpha\beta} Z_{MN} + (\gamma^{M_1 \dots M_5} C^{-1})_{\alpha\beta} Z_{M_1 \dots M_5} \\ + (\gamma^{M_1 \dots M_4} \gamma_{11} C^{-1})_{\alpha\beta} Z_{M_1 \dots M_4} + (\gamma^M \gamma_{11} C^{-1})_{\alpha\beta} Z_M + (\gamma_{11} C^{-1})_{\alpha\beta} Z$$

$$\text{However, from } \gamma^{M_1 \dots M_5} \gamma_{11} = \frac{1}{(10-5)!} (-1)^{5(5-1)/2} \epsilon^{M_1 \dots M_5 M_6 \dots M_{10}} \\ \times \gamma_{M_6 \dots M_{10}}$$

means we can write this up to 5

$$p = 1, 2 \pmod{4} \\ = 1, 2, 5$$

Dim. Reduction:

• Dim-Red on T^n preserves the susy of the theory

• Dim reduced theories possess a new Bosonic Symmetry.

* Suppose our theory is in $(d+1)$:

$$M \cong X \times S^1$$

$$x^\mu, \quad \mu = 0, 1, \dots, d$$

$$Z^{\hat{\mu}} \simeq (x^{\mu}, x_d)$$

$$\mu, \nu: 0, \dots, d-1$$

$$\hat{\mu}, \hat{\nu}: 0, \dots, d$$

$$\bullet X_d' \sim X_d \Rightarrow \underline{X_d' = X_d + 2\pi n R}, n \in \mathbb{Z}$$

$$\Phi(Z^{\hat{\mu}}) = \Phi(x^{\mu}, x_d) = \sum_{k=-\infty}^{\infty} e^{ikx_d/R} \Phi_k(x^{\mu})$$

$$= \Phi_0(x^{\mu}) + \sum_{k \neq 0} e^{ik\Theta} \Phi_k(x^{\mu}) \quad \left| \begin{array}{l} \Theta \equiv \frac{x_d}{R} \\ \Theta \in [0, 2\pi] \end{array} \right.$$

Φ : is any general field: i.e. any spin

- From each field in $(d+1)$ we get an ∞ -# of particles in d -dim.

$$\sum_{k \neq 0} e^{ik\Theta} \Phi_k(x^{\mu}): \text{ yields } \underline{\text{massive terms}}$$

ex: Massless scalar field:

$$\square_{D+1} \phi = 0 = \left(\square_D + \left(\frac{\partial}{\partial x_d} \right)^2 \right) \phi$$

$$\phi = \phi_0(x^{\mu}) + \sum_{k \neq 0} e^{ik \frac{x_d}{R}} \phi_k(x^{\mu})$$

$$\Rightarrow \square_{D+1} \phi_0 = 0$$

$$\left(\square_{D+1} - \left(\frac{k}{R} \right)^2 \right) \phi_k(x^{\mu}) = 0$$

$\rightarrow$ Tower of mass terms

$$\boxed{m_k = \frac{k}{R}} \quad k\text{-th mode / particles.}$$

- In the limit $R \rightarrow 0$, $k \rightarrow \infty$ (unphysical?) and so we ignore these terms

$\phi_0(x^{\mu})$: Zero modes

Metric: $[2000]_g = [2000]_g + [1000]_g + [0000]_g$

$$g_{\hat{\mu}\hat{\nu}} = g_{\mu\nu} + A_{\mu} + e^{\phi}$$

* We write ds_{d+1}^2 in a very particular way

$$ds_{d+1}^2 = e^{2\alpha_d \phi} ds_d^2 + e^{-2(d-2)\alpha_d \phi} (dx_d + A_\mu dx^\mu)^2$$



X_d



ϕ

$[0000]$



A_μ

$[1000]$

$\sim [2000] +$

$$\alpha_d \equiv (2(d-1)(d-2))^{-1/2}$$

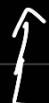
$$= e^{2\alpha_d \phi} g_{\mu\nu} dx^\mu dx^\nu + e^{-2(d-2)\alpha_d \phi} (dx_d)^2 + e^{-2(d-2)\alpha_d \phi} A_\mu dx_d dx^\mu + e^{-2(d-2)\alpha_d \phi} A_\mu A_\nu dx^\mu dx^\nu$$

$$= (e^{2\alpha_d \phi} g_{\mu\nu} + e^{-2(d-2)\alpha_d \phi} A_\mu A_\nu) dx^\mu dx^\nu + e^{-2(d-2)\alpha_d \phi} (dx_d)^2 + e^{-2(d-2)\alpha_d \phi} A_\mu (dx_d dx^\mu + dx^\mu dx_d)$$

$$g_{\hat{\mu}\hat{\nu}} = \begin{pmatrix} e^{2\alpha_d \phi} g_{\mu\nu} + e^{-2(d-2)\alpha_d \phi} A_\mu A_\nu & A_\mu e^{-2(d-2)\alpha_d \phi} \\ A_\mu e^{-2(d-2)\alpha_d \phi} & e^{-2(d-2)\alpha_d \phi} \end{pmatrix}$$

$$g_{\hat{\mu}\hat{\nu}} = e_{\hat{\mu}}^{\hat{a}} e_{\hat{\nu}}^{\hat{b}} \eta_{\hat{a}\hat{b}}$$

$$\Rightarrow e_{\hat{\mu}}^{\hat{a}} = \begin{pmatrix} e^{\alpha_d \phi} e_\mu^m & e^{-(d-2)\alpha_d \phi} A_\mu \\ 0 & e^{-(d-2)\alpha_d \phi} \end{pmatrix}$$



from Lorentz symmetry

$$g_{d\mu} = A_\mu e^{-2(d-2)\alpha_d \phi} = e_d^m e_\mu^n \eta_{mn} = 0?$$

* $0 \leq x_d \leq 2\pi R$, R : has no physical meaning.

$\leadsto 2\pi R_{d+1}^{\text{phys}}$: diffeomorphism invariant distance

$$2\pi R_{d+1}^{\text{phys}} \equiv \int_0^{2\pi R} dx_d \sqrt{g_{dd}} = \int_0^{2\pi R} dx_d e^{-2(d-2)\alpha_d \phi} = 2\pi R e^{-(2-d)\alpha_d \phi}$$

$$R_{d+1}^{\text{phys}} = R e^{-(d-2)\alpha_d \phi}$$

* Einstein term: $\int d^{d+1}x \hat{e} \hat{R}$

$$\hat{e} = e^{-(d-2)\alpha_d \phi} e$$

$$\int d^{d+1}x \hat{e} \hat{R} = 2\pi R \int d^d x e \bar{e}^{-(d-2)\alpha_d \phi} e^{\hat{a}\hat{\mu}} e^{\hat{b}\hat{\nu}} R_{\hat{\mu}\hat{\nu}\hat{\alpha}\hat{\beta}}$$

$$\bullet e^{\hat{a}\hat{\mu}} e^{\hat{b}\hat{\nu}} R_{\hat{\mu}\hat{\nu}\hat{\alpha}\hat{\beta}} = e^{b_d} e^{d\omega} R_{dd\omega\omega}$$

→ Do this calculation separately

$$\int d^{d+1}x \hat{e} \hat{R} = 2\pi R \int d^d x e \left(R - \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{4} e^{-2(d-1)\alpha_d \phi} F_{\mu\nu} F^{\mu\nu} \right)$$

$$\bullet 2\pi R \sim \frac{1}{241} \quad [2000]_8 \oplus [10000]_8 \oplus [1000]_8$$

exactly as we expected.

p-form - dim reduction: $\hat{F}_{\hat{\mu}_1 \dots \hat{\mu}_p} = p \partial[\hat{\mu}_1 \hat{A}_{\hat{\mu}_2 \dots \hat{\mu}_p}]$
define

$$A_{\hat{\mu}_1 \dots \hat{\mu}_{p-2} d} = A_{\mu_1 \dots \mu_{p-2}} \quad (\text{since } A_{dd} = 0)$$

$$\int d^{d+1}x \frac{e}{2p!} F_{(p)}^2 = \int d^d x \int d^d x e^{2\alpha_d \phi} \frac{e}{2p!} F^{\hat{\mu}_1 \dots \hat{\mu}_p} F_{\hat{\mu}_1 \dots \hat{\mu}_p}$$

$$= 2\pi R \int d^d x e^{2\alpha_d \phi} \frac{e}{2p!} F^{\hat{\mu}_1 \dots \hat{\mu}_p} F_{\hat{\mu}_1 \dots \hat{\mu}_p}$$

* Step by step:

$$F^{\hat{\mu}_1 \dots \hat{\mu}_{p-1} d} F_{\hat{\mu}_1 \dots \hat{\mu}_{p-1} d} x p e^{c_1} + F^{\hat{\mu}_1 \dots \hat{\mu}_{p-1} M p} F_{\hat{\mu}_1 \dots \hat{\mu}_{p-1} M p}$$

$$= p e^{(d-2)\alpha_d \phi} F^{\hat{\mu}_1 \dots \hat{\mu}_{p-1} d} F_{\hat{\mu}_1 \dots \hat{\mu}_{p-1} d} + F^{\hat{\mu}_1 \dots \hat{\mu}_{p-1} M p} F_{\hat{\mu}_1 \dots \hat{\mu}_{p-1} M p}$$

$$= p e^{(d-1)\alpha_d \phi} F^{\hat{\mu}_1 \dots \hat{\mu}_{p-1}} F_{\hat{\mu}_1 \dots \hat{\mu}_{p-1}} + F^{\mu_1 \dots \mu_{p-1} M p} F_{\mu_1 \dots \mu_{p-1} M p}$$

①

②

$$(1) p e^{(d-2)\alpha_d \phi} e_{\hat{m}_1}^{m_1} \dots e_{\hat{m}_{p-1}}^{m_{p-1}} F^{m_1 \dots m_{p-1}} F_{m_1 \dots m_{p-1}}$$

$$e_{\hat{m}}^{\hat{m}} = \begin{pmatrix} \binom{M}{m} & \binom{d}{m} \\ \binom{M}{d} & \binom{d}{d} \end{pmatrix}$$

$$\rightarrow e_{\hat{m}}^m \rightarrow \underbrace{e_d^m}_0, e_m^m \quad \binom{u}{m} = e^{-\alpha_d \phi} e_m^u$$

$$= p e^{(d-2)\alpha_d \phi} e^{-\alpha_d \phi (p-1)} e_{m_1}^{m_1} \dots e_{m_{p-1}}^{m_{p-1}} F^{m_1 \dots m_{p-1}} F_{m_1 \dots m_{p-1}} \\ = p e^{(d-p-1)\alpha_d \phi} F_{p-1}^2$$

$$(2) e_{\hat{m}_1}^{m_1} \dots e_{\hat{m}_{p-1}}^{m_{p-1}} e_{\hat{m}_p}^{m_p} F^{\hat{m}_1 \dots \hat{m}_p} F_{m_1 \dots m_p}$$

$$e_d^{m_1} \dots e_{m_p}^{m_p} F^d \dots m_p F_{m_1 \dots m_p}$$

$$= e^{-\alpha_p p \phi} e_{m_1}^{m_1} \dots e_{m_p}^{m_p} F^{m_1 \dots m_p} F_{m_1 \dots m_p} \\ = e^{-p \alpha_p \phi} F_p^2$$

$$(3) e_{\hat{m}_1}^{m_1} \dots e_{\hat{m}_p}^d F^{\hat{m}_1 \dots \hat{m}_p} F_{m_1 \dots m_p}$$

→ still need

$$e^{(p-1)\alpha_d \phi} (e_{m_1}^{m_1} \dots e_{m_{p-1}}^{m_{p-1}}) \underbrace{e_m^d}_{\text{needed}} (-A_\sigma e^{-\alpha_d \phi} e_{m_p}^\sigma) F^{m_1 \dots m_p} F_{m_1 \dots m_p}$$

$$= -e^{-p \alpha_d \phi} e_{m_1}^{m_1} \dots e_{m_p}^{m_p} A_{m_p} F^{m_1 \dots m_p} F_{m_1 \dots m_p}$$

* Overall only doing half really should just work with g_{MN} , or

$$F_{m_1 \dots m_{p-1} d} = e_{m_1}^{\hat{m}_1} \dots e_{m_{p-1}}^{\hat{m}_{p-1}} \underbrace{F_{\hat{m}_1 \dots \hat{m}_{p-1} d}}_1$$

$$p \partial_{[m_1} A_{m_2 \dots m_p]}$$

$$= (p-1) \partial_{[m_1} A_{m_2 \dots m_{p-1}]}$$

$$= e_{m_1}^{m_1} \dots e_{m_{p-1}}^{m_{p-1}} e^{-\alpha_p (p-1) \phi} F_{m_1 \dots m_{p-1}}$$

$$F_{n_1 \dots n_{p-1} d} = e^{-(p-1)\alpha_d \phi} F_{n_1 \dots n_{p-1}}$$

$$\hat{F}_{n_1 \dots n_p} = e_{n_1}^{\hat{v}_1} \dots e_{n_p}^{\hat{v}_p} F_{\hat{v}_1 \dots \hat{v}_p}$$

$$= p e_{n_1}^{\hat{v}_1} \dots e_{n_p}^{\hat{v}_p} F_{\hat{v}_1 \dots d} (-A_{\nu} e^{-\alpha_d \phi} e_{n_p}^{\hat{v}_p}) \times e^{-\alpha_d \phi (p-1)} + \underbrace{e_{n_1}^{\hat{v}_1} \dots e_{n_p}^{\hat{v}_p}}_{e^{-\alpha_d (p-1) \phi}} F_{\hat{v}_1 \dots n_p}$$

$$= e^{-p\alpha_d \phi} (F_{n_1 \dots n_p} - p F_{n_1 \dots n_{p-1}} A_{n_p}) \quad \checkmark$$

$$F_p^2 \rightarrow e^{-\alpha_d (2p-2d+2)\phi} F_{p-1}^2 = e^{2\alpha_d (d-p)\phi} e^{-2\alpha_d \phi}$$

* Try $p-1$ term again: $e_{\hat{v}_1}^{\hat{n}_1}$

$$2\pi R \int d^d x \frac{e^{2\alpha_d \phi}}{2p!} p F_{\hat{n}_1 \dots \hat{n}_{p-1} d} F_{\hat{n}_1 \dots d}$$

$$= 2\pi R \int d^d x \frac{e^{2\alpha_d \phi}}{2p!} p F_{n_1 \dots d} e^{\alpha_d \phi (p-1)}$$

$$= \int d^d x \frac{e^{2\alpha_d \phi}}{2p!} p e^{\alpha_d \phi (p-1)} e^{2\alpha_d (d-p-1)\phi} F_{p-1}^2$$

Ohhh ... $\det e_{\hat{n}}^{\hat{v}} = \begin{vmatrix} e^{\alpha_d} e_m^m & A_{..} \\ 0 & e^{-(d-2)\alpha_d \phi} \end{vmatrix}$

for $e^{\alpha_d \phi} e_m^m$ we get d terms i.e.

$$e^{d\alpha_d \phi} \cdot e^{(-d+2)\alpha_d \phi} = e^{2\alpha_d \phi}$$

$$e_{d+1} = e^{2\alpha_d \phi} e_d$$

$$= 2\pi R \int d^D x \frac{e^{-2(p-1)\alpha\phi}}{2(p-1)!} F_{(p-1)}^2 \quad (\delta\text{-term})$$

• other term:

$$= \frac{F^{\mu_1 \dots \mu_{p-1} \mu_p} F_{\mu_1 \dots \mu_{p-1} \mu_p}}{e^{-p\alpha\phi} p!}$$

$\hat{g}_{\mu\nu} \rightarrow g_{\mu\nu}$, n -vectors $g_{\mu i}$, g_{ij} $\frac{n(n-1)}{2}$ scalars ϕ_i , $i = 1 \dots n$

• grav-photon: $g_{\mu\nu}$

$A_{\hat{\mu}_1 \dots \hat{\mu}_p}$: scalars $A_{i_1 \dots i_p}$

* Torus comp:

$$\int d^D x \frac{e^{-2\alpha\phi}}{2p!} \vec{B} \cdot \vec{\Phi} F_{(p)}^2$$

• for $p+1$ dim: $\int d^D x \frac{e^{-2\alpha\phi}}{2p!} \vec{B} \cdot \vec{\Phi} F_{(p)}^2$ Aux field

an equivalent action: $\int d^D x \left(\frac{e^{-2\alpha\phi}}{2p!} \vec{B} \cdot \vec{\Phi} F_{(p)}^2 + \frac{e^{-2\alpha\phi}}{p!} \varphi \partial_\mu F_{\nu_1 \dots \nu_p} \epsilon^{\mu\nu_1 \dots \nu_p} \right)$

$F_{\mu \dots}$: thought to be an independent field.

φ : Lagrange multiplier

E.O.M $\rightarrow \varphi$: $\partial_\mu F_{\nu_1 \dots \nu_p} \epsilon^{\mu\nu_1 \dots \nu_p} = 0$

$d \star F = 0$

$\star F = dG$

$\Rightarrow - \int d^D x \partial_\mu \varphi \frac{e^{-2\alpha\phi}}{p!} F_{\mu\nu_1 \dots \nu_p} \epsilon^{\mu\nu_1 \dots \nu_p}$

E.O.M:

$$\left(\frac{e^{-2\alpha\phi}}{p!} \vec{B} \cdot \vec{\Phi} F^{\mu_1 \dots \mu_p} - \frac{e^{-2\alpha\phi}}{p!} \partial_\mu \varphi \epsilon^{\mu\nu_1 \dots \nu_p} \right) = 0$$

$\therefore$ $F^{\mu_1 \dots \mu_p} = e^{-2\alpha\phi} \partial_\mu \varphi \epsilon^{\mu\nu_1 \dots \nu_p}$ $F_{\mu\nu}$ E.O.M.

Plug back into $\int d^D x$:

$$\int d^D x e^{-2\alpha\phi} \frac{\partial_\mu \varphi \partial^\sigma \varphi \epsilon^{\mu\nu_1 \dots \nu_p} \epsilon_{\sigma\nu_1 \dots \nu_p}}{2p!} e^{-2\alpha\phi}$$

$$= \int d^{p+1}x \frac{e}{p!} \partial_\mu \varphi \left(e^{-\vec{B} \cdot \vec{\Phi}} \xi_{\mu \nu_1 \dots \nu_p} \partial^{\nu_1} \varphi \right) \xi^{\mu \nu_1 \dots \nu_p}$$

$$\xi_{\sigma \nu_1 \dots \nu_p} \xi^{\mu \nu_1 \dots \nu_p} = -p! \delta_\sigma^\mu$$

$$= -\frac{1}{2} \int d^{p+1}x e^{-\vec{B} \cdot \vec{\Phi}} \partial_\mu \varphi \partial^\mu \varphi e + \int d^{p+1}x e^{-\vec{B} \cdot \vec{\Phi}} \partial_\mu \varphi \partial^\mu \varphi e$$

$$\therefore = \int d^{p+1}x \frac{e}{2} e^{-\vec{B} \cdot \vec{\Phi}} \partial_\mu \varphi \partial^\mu \varphi \quad \left(\text{which describes a scalar} \right)$$

* In 3D - we can dualize all 2-forms in terms of scalars.

$$A_{(2)} \Rightarrow * dA^{(2)} = * B^{(3)} = B^{(1)}$$

↓ Compactify down to 3D, all terms from $\int R e$

$$\int d^4x e \left(R_4 - \frac{1}{2} \partial_\mu \varphi \partial^\mu \varphi - \frac{1}{4} e^{-2G-1} \det \phi F_{\mu\nu} F^{\mu\nu} \right)$$

$$\downarrow$$

$$\int d^3x e \left(R_3 - \frac{1}{2} \partial_\mu \vec{\Phi} \cdot \partial^\mu \vec{\Phi} - \frac{1}{2} \sum_{\vec{B}} e^{-\vec{B} \cdot \vec{\Phi}} \partial_\mu \chi_{\vec{B}} \partial^\mu \chi_{\vec{B}} + \dots \right)$$

Note: Metrics differ

(i) String frame $g_{\mu\nu} = e^{\phi/2} g_{\mu\nu}^E$

(ii) Einstein frame

$$\vec{\Phi} = (\phi_1, \dots, \phi_n), \quad \vec{B} \in \mathbb{Z}^n, \quad \chi_{\vec{B}}:$$

$\vec{\Phi}$: arise from diagonalization of $g_{\mu\nu}$

$\chi_{\vec{B}}$: arise from off-diagonal components of $g_{\mu\nu}/A_\mu$

• Some from dualization.

→ Each have different symmetries

$\chi_{\vec{B}}$: $g_{cd} \oplus$ gauge appear: $\partial_\mu \chi_{\vec{B}}$

$\vec{\Phi}$: Non derivative interactions

• Ignores C.S. terms, if we incorporate this term:

$$\partial_\mu \chi_{\vec{\alpha}} \rightarrow \underbrace{A_{\vec{\alpha}}^{\vec{B}}}_{\text{non-deg matrix}} \partial_\mu \chi_{\vec{B}}$$

* Scalars and Vacuum states:

In S.M. we have a single scalar (Higgs) field which determines the vacuum state $\langle 0 | \phi | 0 \rangle$.

→ Thus, we should count the scalars in our theory:

$$\underline{D=11}: \quad \left(\underbrace{e_\mu^M, A^{(3)}}_{\text{Boson Sector}}, \underbrace{\psi_\mu^a}_{\text{Ferm. Sector}} \right)$$

$$M^{1,1} = \Sigma^3 \times T^8$$

11D sugra on T^8 down to a 3-manifold.

$$\begin{array}{c} g_{\mu\nu}^{\Sigma^3} + A^{(3)} \\ \downarrow \qquad \downarrow \end{array}$$

$$36 + 56 = \underline{92}$$

Scalars

$$g_{ij} \quad A_{ijk}, \quad i, j, k = 1, \dots, 8$$

• But when we dualize $g_{\mu i}, A_{\mu ij}$ we get

$$8 + 28 = \underline{36}$$

new scalars 128 in 3D

* Scalars belong to the non-linear realization of

E_8 with local Subgroup $SO(16) \subset E_8$

$$|E_8/SO(16)| = \underline{128}$$

Back to theory:

$D=11$

$e_m^{\hat{m}}$

$A^{(3)}$

$\psi_{\hat{m}\alpha}$

$\downarrow$

$\downarrow$

$\downarrow$

$e_m^M, B_{m\sigma}$

$A_{M\sigma\sigma_2}, A_{M\sigma\sigma_2}$

$\psi_{m\alpha}, \lambda_\alpha$

$\phi = -\sigma$

$$e_{\hat{m}}^{\hat{m}} = \begin{pmatrix} e^{-\frac{1}{12}\sigma} e_m^M & e^{\frac{2}{3}\sigma} A_m \\ 0 & e^{\frac{2}{3}\sigma} \end{pmatrix}, (e^{-1})_{\hat{m}}^{\hat{m}} = \begin{pmatrix} e^{\frac{1}{12}\sigma} e_m^M & -e^{\frac{1}{12}\sigma} A_m \\ 0 & e^{\frac{2}{3}\sigma} \end{pmatrix}$$

$B_m = \frac{1}{2} A_m$

* Definition: $\bar{\psi}_{\hat{m}} \gamma^{\mu\nu\rho} D_\nu (\frac{1}{2}(\omega + \tilde{\omega})) \psi_{\hat{m}} \big|_{\alpha_{10}} = \left(\frac{1}{29.8} \right)^{1/2} = \frac{1}{12}$

• $\psi_{\hat{m}} e_m^{\hat{m}}$

$$\begin{aligned} \psi_{\hat{m}} e_m^{\hat{m}} &= \psi_{\hat{m}} \tilde{e}_m^{\hat{m}} + \psi_{\hat{d}} \tilde{e}_m^{\hat{d}} \\ &= \psi_{\hat{m}} e^{\frac{1}{12}\sigma} e_m^M - \psi_{\hat{d}} e^{\frac{1}{12}\sigma} A_m \\ &= e^{\frac{1}{12}\sigma} (\psi_{\hat{m}} - A_m \psi_{\hat{d}}) = \underline{e^{\frac{1}{12}\sigma} \psi'_{\hat{m}}} \end{aligned}$$

$$\begin{aligned} \psi_{\hat{m}} \tilde{e}_{10}^{\hat{m}} &= \cancel{\psi_{\hat{m}} \tilde{e}_{10}^{\hat{m}}} + \psi_{10} \tilde{e}_{10}^{\hat{10}} \\ &= \underline{e^{-\frac{2}{3}\sigma} \psi_{10}} \end{aligned}$$

$$\tilde{\psi}_{\hat{m}} \rightarrow \begin{cases} \psi_{\hat{m}} \\ \lambda_\alpha \end{cases}$$

$$\psi_{10} = \frac{2\sqrt{2}}{3} e^{\frac{17}{24}\sigma} \lambda$$

$$\therefore \psi_{\hat{m}} \tilde{e}_{10}^{\hat{m}} = \frac{2\sqrt{2}}{3} e^{\frac{17}{24}\sigma - \frac{16}{24}\sigma} \lambda = \frac{2\sqrt{2}}{3} e^{\frac{1}{24}\sigma} \lambda$$

* From: $\psi_{\hat{m}} \tilde{e}_{\hat{m}}^{\hat{m}} = e^{\frac{1}{12}\sigma} (\psi_{\hat{m}} - A_m \frac{2\sqrt{2}}{3} e^{\frac{17}{24}\sigma} \lambda)$

$$\boxed{\tilde{\psi} = e^{-\frac{1}{24}\sigma} \psi} \bigg| A_m \frac{2\sqrt{2}}{3} e^{\frac{17}{24}\sigma} = \frac{e^{-\frac{1}{24}\sigma}}{6\sqrt{2}} \gamma_{\hat{m}} \gamma^{\hat{m}}$$

$$A_m \propto e^{16/24\sigma} = \gamma_m \gamma''$$

???

$$\Rightarrow A_m \equiv 8 e^{-16/24\sigma} \gamma_m \gamma'' \quad ???$$

* Q: How does A_m , coming from $\hat{E}_m^{\hat{m}}$ yield an element of $\text{Spin}(2,9)/\mathbb{Z}_2$?

$$\gamma_m \gamma'' = \begin{pmatrix} 0 & \sigma^m \\ \bar{\sigma}^m & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -\sigma^m \\ \bar{\sigma}^m & 0 \end{pmatrix}$$

$$\lambda = P_L \lambda + P_R \lambda$$

I'm assuming this comes from triality of $SO(8)$ between $[0000] \leftrightarrow [0010] \leftrightarrow [0001]$

$$\psi'_m \propto e^{-\frac{\sigma}{24}} \left(\psi_m - \frac{1}{6\sqrt{2}} \gamma_m \gamma'' \lambda \right)$$

$$\therefore L_{\text{IIA}} = L^B + L^F$$

* Last: Dim red of C.S. term:

$$\frac{2k_{11}}{(12)^4} e^{\hat{m}_1 \dots \hat{m}_7} F_{\hat{m}_0 \dots \hat{m}_3} F_{\hat{m}_4 \dots \hat{m}_7} A_{\hat{m}_8 \hat{m}_9 \hat{m}_{10}}$$

$$F^{(4)} \rightarrow \tilde{F}^{(3)} e^{-\frac{5}{12}\sigma}$$

$$A^{(3)} \rightarrow A^{(2)} ?$$

$$A_{\hat{m}_1 \dots \hat{m}_{p-1} \hat{d}} = e^{\hat{m}_1} e^{\hat{m}_{p-1}} A_{\hat{m}_1 \dots \hat{m}_{p-1}}$$

$$\begin{aligned} \text{but } \int F^{(3)} \wedge F^{(4)} \wedge A^{(3)} &= \alpha \int dA^{(2)} \wedge F^{(4)} \wedge A^{(3)} \\ &+ \beta \int F^{(4)} \wedge dA^{(2)} \wedge A^{(3)} \\ &+ \gamma \int F^{(4)} \wedge F^{(4)} \wedge A^{(2)} \end{aligned}$$

$$\begin{aligned} &= \alpha (-1)^6 \int A^{(2)} \wedge F^{(4)} \wedge F^{(4)} \\ &+ \beta (-1)^2 \int F^{(4)} \wedge A^{(2)} \wedge F^{(4)} \end{aligned}$$

$$\begin{aligned}
 & + \gamma \int F^{(4)} \wedge F^{(4)} \wedge A^{(2)} \\
 & = (\alpha' + \beta' + \gamma') \int F^{(4)} \wedge F^{(4)} \wedge A^{(2)} \\
 & = \frac{1}{2 \times (12)^2} \epsilon^{\mu_1 \dots \mu_{10}} F_{\mu_1 \dots \mu_4} F_{\mu_5 \dots \mu_8} A_{\mu_9 \mu_{10}}
 \end{aligned}$$

$$\therefore L^B = \frac{1}{2\kappa_0^2} \left[(\text{E.m.s.}) - (F^2) + (\text{C.S}) \right]$$

$$\begin{aligned}
 L^B = \frac{1}{2\kappa_0^2} & \left[e R_{10} - \frac{e}{2} \partial_\mu \phi \partial^\mu \phi - \frac{e}{4} e^{\frac{3}{2}\sigma} F_{\mu_1 \mu_2} F^{\mu_1 \mu_2} \right. \\
 & - 4 \left(\frac{e}{48} e^{\frac{1}{2}\sigma} \hat{F}_{\mu_1 \dots \mu_4} \hat{F}^{\mu_1 \dots \mu_4} + \frac{e}{12} e^{-\sigma} F_{\mu_1 \mu_2 \mu_3} F^{\mu_1 \mu_2 \mu_3} \right) \\
 & \left. + \left(\frac{1}{2(12)^2} \epsilon^{\mu_1 \dots \mu_{10}} F_{\mu_1 \dots \mu_4} F_{\mu_5 \dots \mu_8} A_{\mu_9 \mu_{10}} \right) \right]
 \end{aligned}$$

* Missing a factor of 4 from $\hat{F}_{(4)}^2$ term.
where

$$\begin{aligned}
 F_{\mu_1 \mu_2} &= 2 \partial_{[\mu_1} A_{\mu_2]} \\
 F_{\mu_1 \mu_2 \mu_3} &= 3 \partial_{[\mu_1} A_{\mu_2 \mu_3]} \\
 \hat{F}_{\mu_1 \dots \mu_4} &= 4 \left(\partial_{[\mu_1} A_{\mu_2 \dots \mu_4]} + A_{[\mu_1} F_{\mu_2 \mu_3 \mu_4]} \right)
 \end{aligned}$$

* L^F given in text.

* Susy transf. found from the dim ansatz for the field transf. in 11d:

$$\delta \hat{e}_{\hat{\mu}}^{\hat{\mu}} = k_{\mu} \bar{\epsilon} \gamma^{\mu} \psi_{\mu}$$

$$\delta \hat{e}_{\hat{\mu}}^{\hat{\mu}} = k_{\mu} \bar{\epsilon} \gamma^{10} \hat{\psi}_{\hat{\mu}}$$

$$\delta \hat{e}_{\hat{\alpha}}^{\hat{\alpha}} = k_{\mu} \bar{\epsilon} \gamma^{\mu} \frac{2\sqrt{2}}{3} e^{\frac{17}{24}\sigma} \chi \quad ?$$

$$\delta e_{\mu}^{\mu} = e \gamma^{\mu} \psi_{\mu}$$

try for

$$\delta e_{\mu}^{\mu} = e \gamma^{\mu} \psi_{\mu}$$

$$\delta \hat{e}_{\hat{\mu}}^{\hat{\mu}} = \dots$$

$$\begin{aligned} \delta \epsilon_\mu &= \delta(e^{3\sigma}) A_\mu + e^{3\sigma} \delta A_\mu \\ &= \epsilon \gamma^\mu \psi_\mu = \epsilon \gamma^\mu \left(e^{\frac{1}{2\alpha\sigma}} \left(\psi_\mu - \frac{1}{6\sqrt{2}} \gamma_\mu \gamma'' \lambda \right) \right) \\ \frac{3\sigma}{12} &= \frac{1}{24} \quad \frac{3\sigma}{24} \end{aligned}$$

Susy transformations:

$$\delta \psi_\mu = \frac{1}{k_\mu} D_\mu(\hat{\Omega}) \epsilon + \frac{1}{12} (\gamma_{\hat{\mu}}^{N_1 N_2 N_3 N_4} - 8 \delta_{\hat{\mu}}^{N_1} \gamma^{N_2 N_3 N_4}) \hat{F}_{N_1 N_2} \epsilon$$

Becker:

$$\delta \psi_\mu = D_\mu^{(1)} \epsilon + \frac{1}{12} (\Gamma_\mu \frac{1}{4!} F_{NPQR} \Gamma^{NPQR} - 3 \frac{1}{3!} (F_{MNPQ} \Gamma^{MNPQ})) \epsilon$$

$$D_\mu(\omega) \epsilon = \partial_\mu \epsilon + \frac{1}{4} \omega_{\mu AB} \Gamma^{AB} \epsilon$$

* Drop all second order ψ terms

$$\omega_{\mu AB} = \frac{1}{2} (-C_{\mu AB} + C_{AB\mu} - \Omega_{B\mu A})$$

$$C_{\mu\nu A} = 2 \partial_{[\mu} E_{\nu]}^A \quad (\text{unhol coordinates})$$

-> Becker ignores fermi terms...

$$\begin{aligned} \delta \psi_A &= e_A^\mu \partial_\mu \epsilon + \frac{1}{4} e_A^\mu \omega_{\mu BC} \Gamma^{BC} \epsilon \\ &\quad + \frac{1}{24} (3 F^{(4)} \Gamma_A - \Gamma_A F^{(4)}) \epsilon \end{aligned}$$

$$\omega_{abc} \Gamma^{BC} = 2 \omega_{a[bc} \Gamma^{c]} + \omega_{abc} \Gamma^{bc}$$

$$\text{Use } \omega = -\frac{1}{2} (-C + e - C)$$

$$\therefore \omega_{a[bc} \Gamma^{BC} = e^{\phi/3} (\omega_{abc} \Gamma^{bc} - \frac{2}{3} \Gamma_a^\mu \partial_\mu \phi) + e^{4\phi/3} F_{ab} \Gamma^b \Gamma_{11}$$

$$F_{ab} = 2 \partial_{[a} A_{b]}$$

$$\omega_{a[bc} \Gamma^{BC} = -\frac{1}{2} e^{4\phi/3} F_{bc} \Gamma^{bc} - \frac{4}{3} e^{\phi/3} \Gamma_a^\mu \Gamma_{11} \partial_\mu \phi$$

Thus:

$$\delta \psi_{11} = E_{11}^\mu \partial_\mu \epsilon + \frac{1}{4} \omega_{12BC} \Gamma^{BC} \epsilon + \frac{1}{24} (3 F^{(4)} \Gamma_{11} - \Gamma_{11} F^{(4)}) \epsilon$$

since $E_{11}^\mu = 0$ for Becker (opposite to our name!)

$$F_{abcd}^{(1)} = e^{4\phi/3} (F_{abcd} + 4\Lambda_{Ta} H_{bcd}) = e^{4\phi/3} \tilde{F}_{abcd}$$

$$F_{abc11}^{(1)} = e^{4\phi/3} H_{abc}$$

$$F^{(4)} = e^{4\phi/3} \tilde{F}^{(4)} + e^{4\phi/3} H^{(3)} \Gamma_{11}$$

$$\delta\psi_{11} = \frac{1}{4} \left[-\frac{1}{2} e^{4\phi/3} F_{ab} \Gamma^{bc} - \frac{4}{3} e^{4\phi/3} \Gamma^\mu \Gamma_{11} \partial_\mu \phi \right] \epsilon$$

$$+ \frac{1}{24} \left[3 \left(e^{4\phi/3} \tilde{F}^{(4)} + e^{4\phi/3} H^{(3)} \right) \Gamma_{11} \right. \\ \left. - \Gamma_{11} \left(e^{4\phi/3} \tilde{F}^{(4)} + e^{4\phi/3} H^{(3)} \right) \right] \epsilon$$

$$\delta\psi_{11} = (e^{4\phi/3}) \left(-\frac{1}{4} e^\phi F^{(2)} \right) \epsilon - \frac{1}{3} e^{4\phi/3} \partial_\mu \phi \Gamma^\mu (\Gamma_{11} \epsilon)$$

$$+ \frac{e^{4\phi/3}}{24} \left(3 \tilde{F}^{(4)} (\Gamma_{11}) - \Gamma_{11} \tilde{F}^{(4)} \right) \epsilon$$

$$+ \frac{e^{4\phi/3}}{24} \left(3 H^{(3)} \Gamma_{11} - \Gamma_{11} H^{(3)} \right) \epsilon$$

D = Even

$$\{ \gamma_{11}, \gamma^{odd} \} = 0$$

$$[\gamma_{11}, \gamma^{even}] = 0$$

$$e^{-4\phi/3} \delta\psi_{11} = -\frac{1}{4} e^\phi F^{(2)} \epsilon - \frac{1}{3} \partial_\mu \phi \Gamma^\mu (\Gamma_{11} \epsilon)$$

$$+ \frac{1}{12} \tilde{F}^{(4)} \Gamma_{11} \epsilon + \frac{1}{6} H^{(3)} \Gamma_{11} \epsilon$$

and likewise for

$e^{-4\phi/3} \delta\psi_a$, and we redefine fields and obtain the susy terms

Symmetry of Type IIA:

* Action: has an $so(1,1)$ symmetry w parameter c :

$$\left\{ \begin{aligned} \sigma &\rightarrow \sigma + c, \quad A_\mu \rightarrow e^{-3c/4} A_\mu, \quad A_{\mu\nu} \rightarrow e^{c/2} A_{\mu\nu} \\ A_{mnp} &\rightarrow e^{-c/4} A_{mnp} \end{aligned} \right\}$$

• $e^{\frac{c}{12}}$ insert

→ origin from the I1D theory: invariant under:

$$e_m^{\mu} \rightarrow e^{-\alpha} e_m^{\mu}, \quad \psi_m \rightarrow e^{\alpha/2} \psi_m$$

$$A_{\mu_1 \mu_2 \mu_3} \rightarrow e^{-3\alpha} A_{\mu_1 \mu_2 \mu_3}$$

so long as $K_{11} \rightarrow e^{-9\alpha/2} K_{11}$

* Take: $\boxed{\alpha = c/12} \mid x^{10} \rightarrow e^{-\alpha} x^{10}$

$$(i) A_{\mu\nu 10} \rightarrow A'_{\mu'\nu' 10'} = \left(\frac{\partial x^{10}}{\partial x^{10'}} \right) A'_{\mu\nu}$$

$$(x^{10})' = e^{-9\alpha} x^{10} = (e^{9\alpha}) (e^{-3\alpha} A_{\mu\nu})$$

$$A'_{\mu\nu} \rightarrow e^{6\alpha} A_{\mu\nu} = e^{c/2} A_{\mu\nu}$$

(as claimed!)

$$\boxed{e_{10}^{10} = e^{\frac{2}{3}\sigma}}$$

(ii) Lets try for σ :

$$e_{10}^{10} = e^{2\sigma/3}$$

$$e_{10}^{10} \rightarrow e^{-\alpha} (e_{10'}^{10}) = e^{-\alpha} \frac{\partial x^{10}}{\partial x^{10'}} e_{10}^{10}$$

$$\begin{aligned} e^{-\alpha} e^{2\sigma/3} e^{9\alpha} &= e^{9\alpha - \alpha} e^{8\alpha} e_{10}^{10} \\ &= e^{8\alpha} e_{10}^{10} \\ &= e^{8/12 c} e^{2\sigma/3} \\ &= e^{\frac{2}{3}c} e^{\frac{2}{3}\sigma} \\ &= \exp\left(\frac{2}{3}(c + \sigma)\right) \end{aligned}$$

$$\sigma \rightarrow c + \sigma$$

* Scales the Lagrangian by

$$L \rightarrow e^{9\alpha} L$$

and so we rescale K_{11} as claimed.

• Now, in 10D:

$$x^{10} \rightarrow e^{-9\alpha} x^{10} = (x^{10})'$$

$$\int dx^{10} \rightarrow e^{-9\alpha} \int dX^{10}$$

which cancels out the above $e^{9\alpha}$ transformation
 * (Weyl in 10D) $\oplus$ ($X^{10} \rightarrow e^{9\alpha} X^{10}$) leaves
 $\int \mathcal{L}$ invariant.

$$\text{In 10D: } \left(\int dx^{10} e^{10}_0 \right) \rightarrow \int dX^{10} e^{-9\alpha} e^{9\alpha} e^{20/3} \\ = 2\pi R e^{20/3}$$

* $\left(\frac{2\pi R_{11}}{K_{11}^2} \right)$: defines a 10D Newton coupling const.

$$K_{10}^{-2} = 2\pi R_{11} K_{11}^{-2}$$

* To keep X^{10} the same we take $R \rightarrow e^{-9\alpha} R$
 $K_{11}^{-2} \rightarrow e^{9\alpha} K_{11}^{-2}$

$\rightarrow K_{10}^{-2}$ is invariant and scaling in the 10D theory is required.

Type IIA: (string perspective)

$\rightarrow$ Sigma model: 2D string action

$$S_s = - \frac{1}{4\pi\alpha'} \int d^2\xi \sqrt{-g} g^{\alpha\beta} \left(\partial_\alpha X^\mu \partial_\beta X^\nu g_{\mu\nu}^S - \epsilon^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu A_{\mu\nu} \right) \\ + \frac{1}{4\pi} \int d^2\xi \sigma R^{(2)} + (\text{fermions})$$

$\rightarrow$ String coupled to $(g_{\mu\nu}^S, A_{\mu\nu}, \sigma)$

$\left\{ \begin{array}{l} g_{\mu\nu}^S: \text{metric} \\ A_{\mu\nu}: \text{2-form Gauge field} \\ \sigma: \text{Dilaton} \end{array} \right.$

* all occur in the massless spectrum of the

string.

• Background fields $X^M(\tau, \sigma)$

$$[\alpha'] \sim (\text{mass})^{-2}$$

$$T_1 = \frac{1}{2\pi\alpha'} \quad (\text{String tension})$$

→ Demand this theory to be conformally invariant as a QFT.

$$\rightarrow \beta_{\alpha'} (\text{Beta fns}) \stackrel{!}{=} 0$$

• Recover tree-level string scattering egs to lowest order in α'

→ Remarkably recover the supergravity E.C.M.

$$* \text{Eins. action: } \sim e^{-2\sigma} eR$$

$$e^{SM} = e^{1/4\sigma} e_m^m$$

eR: Einstein frame

$$\text{string frame: } e_m^{SM} = e^{0/4\sigma} e_m^m$$

→ 13.4.14: does not contain the massless fields of Π/A s.l.

$$\rightarrow \text{Change } e_m^m = e^{-\frac{\sigma}{4}} e_m^{SM}$$

$$\begin{aligned} \mathcal{L}^B = \frac{1}{2\alpha'^2} \left\{ e e^{-2\sigma} \left[R + 4 \partial_\mu \sigma \partial^\mu \sigma - \frac{1}{3} F_{\mu_1 \mu_2 \mu_3} F^{\mu_1 \mu_2 \mu_3} \right] \right. \\ \left. + \left[-\frac{1}{12} e F'_{\mu_1 \dots \mu_4} F'^{\mu_1 \dots \mu_4} - \frac{1}{4} e F_{\mu_1 \mu_2} F^{\mu_1 \mu_2} \right] \right. \\ \left. + \frac{1}{2 \times 12^2} e^{\mu_1 \dots \mu_{10}} F_{\mu_1 \dots \mu_4} F_{\mu_5 \dots \mu_8} A_{\mu_9 \mu_{10}} \right\} \end{aligned}$$

→ Form in Becker: (String frame)

* Type IIA SUGRA: low energy limit of type IIA string theory.

(Type IIA String theory)

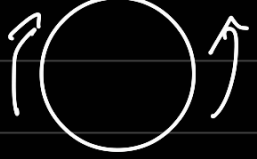
$$\downarrow \begin{matrix} \alpha' \rightarrow \infty \\ T' \rightarrow 0 \text{ (tensionless)} \end{matrix}$$

(Type IIA Supergravity) (Massless)

* Understanding the type IIA SUGRA spectra in terms of type IIA string states:

Type IIA: Closed-string

Type IIB: Open-string

IIA:  L, R: movers, so we have spinors of opposite chirality

* NS-R string: which has 4 sectors

$(NS \oplus R \oplus R \oplus NS)$ (fermions) $\oplus$ $(NS \oplus NS \oplus R \oplus R)$ (bosons)

→ Are all Boundary conditions on the fermions

$$\int d^2\xi e \left\{ -\frac{1}{2} e_a^\alpha \partial_\alpha x^\mu e^{\alpha\beta} \partial_\beta x^\mu - \frac{i}{2} \bar{\chi}^\mu \not{\partial} \chi^\mu + \frac{1}{2} N^\mu N^\mu \right. \\ \left. + \frac{i\kappa}{2} \bar{\Psi}_\alpha \not{\partial} x^\mu \gamma^\alpha \chi^\mu - \frac{\kappa^2}{16} \bar{\Psi}_\alpha \gamma^\beta \gamma^\alpha \Psi_\beta \bar{\chi}^\mu \chi^\mu R_{\mu\nu} \right\}$$

↓ (Quantize $\oplus$ GSO projections)

$e_\mu, A_{\mu\nu}, \sigma, A_{\mu\nu\rho}, A_\mu, \psi_{\mu\alpha}, \lambda_\alpha$

$NS \otimes NS$	$R \otimes R$	$NS \otimes R$	$R \otimes NS$
$[2000]_8$	$[0011]_8$	$[1010]_8$	$[0001]_8$
$[0100]_8$	$[1000]_8$	$[1001]_8$	$[0010]_8$
$[0000]_8$			
↑			
(gravity invariant)			
$SO(8)$			

* In the string frame:

(i) $NS \otimes NS$: occur with a factor of $e^{-2\sigma}$ related to the fact that the fields from $R \otimes R$ sector do not appear in the 2D string action

• Type IIA Supergravity has two parameters

- (i) κ_{10}^2 : Gravitational Coupling.
- (ii) $e^{\phi/2}$

• Type IIA string has

- (i) String tension $T' = \frac{1}{2\pi\alpha'}$
- (ii) g_s : String coupling

$\therefore$ these two sets of couplings must be related

Stringy IIA:

$$\int (\dots) e^{g_s^{-2} S_s}$$



g_s : appears in front of cubic vertex

$$1/\alpha' = N-2 \text{ } \dots$$

$$A(1 \dots n) = g_s \downarrow \downarrow \dots$$

$$\Delta(123) = g_s^2 \downarrow \downarrow$$

- Can redefine fields to absorb g_s
- α' : only appears in the masses of the particles

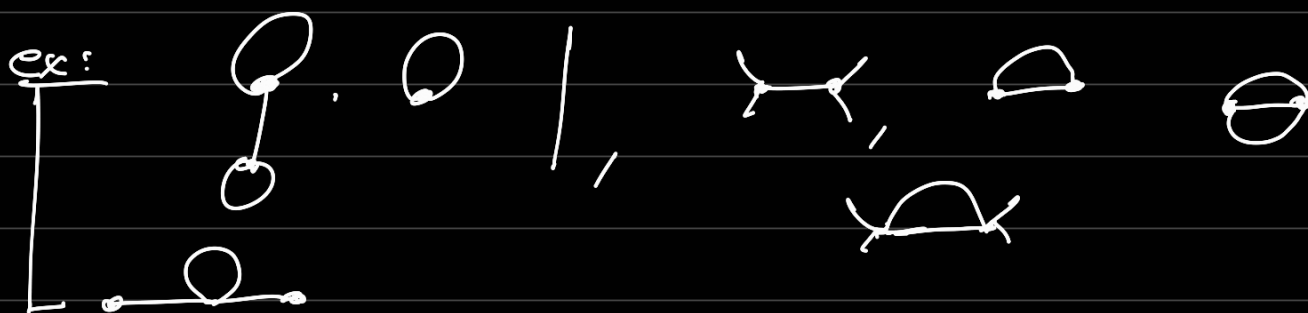
L_0 : massless modes

L_n : massive states $\sim \alpha'$

Path integral: $e^{iS} \rightarrow (i g_s^2)^{-1}$

- So $\hbar$ and g_s occur only as a combination $\hbar g_s^2$

$h^{(n)}$: (n) measures the number of loops.



* Feyn graph with n -loops, I propagators and V vertices

$$\hbar^{(I-V)}$$

• Vertex: $\hbar$		$n = I - V + 1$
• Propagator: $\hbar^{-1}$		

$$\hbar^{(n-1)} = \hbar^{(I-V)}$$

ex: $n = 1 - 2 + 1 = 0$

Note: n is internal loops

$$\begin{array}{l} \text{---} \bigcirc \text{---} \quad n = 3 - 2 + 1 = 2 \quad \checkmark \\ \text{---} \bigcirc \text{---} \quad n = 2 - 2 + 1 = 1 \quad \checkmark \end{array}$$

$\therefore$ n-loop: g_s^{2n-2}

← Euler #

$$\# \int d^2 \xi \sigma R^{(2)} = \langle \sigma \rangle \chi$$

$$\chi = (2 - 2g)$$

→ Each hole in Σ^g is a string loop and hence

$$g = n$$

$$\int \frac{Dx D\gamma}{V(\text{Diff World})} e^{-\tilde{h}' \cdot S} (e^{-\langle \sigma \rangle (2-2n)})$$

$$= \int D\tilde{x} e^{-S[\tilde{x}, \gamma, \dots]} (e^{\langle \sigma \rangle})^{2n-2}$$

$\therefore$ $g_s = e^{\langle \sigma \rangle}$ (String Coupling)

* Massless, low energy: $\alpha' \rightarrow \infty$, tree-level

$\therefore n=0 = g$, action is mult. by

$$e^{-2\langle \sigma \rangle} (1 + \tilde{h}' \cdot S)$$

so we get terms $e^{-2\langle \sigma \rangle} e^{\tilde{h}' \cdot S}$

which we find above for the NS $\bar{c}$ NS sector.

$$[K_{10}] = (\text{mass})^{-4} \propto (\alpha')^2$$

• Since we said $[\alpha'] = (\text{mass})^{-2}$

$$[\text{mass}] = \frac{1}{\alpha'} \Rightarrow [\alpha'] = (\text{mass})^{-2}$$

$$2K_{10}^{-2} = (2\pi)^7 (\alpha')^4 e^{-2\phi}$$

* For IIA: A_m , A_{mn} and A_{mnp}

1 2 3

$F^{(2)}$ $F^{(3)}$ $F^{(4)}$

$$* F^{(2)} = G^{(8)} = d B^{(7)}$$

$$* F^{(3)} = G^{(7)} = d B^{(6)}$$

$$* F^{(4)} = G^{(6)} = d B^{(5)}$$

$$G_{\mu_1 \dots \mu_{D-2-1}} = \frac{1}{(q+1)!} \epsilon_{\mu_1 \dots \mu_{10}} F^{\mu_0-1 \dots \mu_{10}} \Delta^n(M)$$

Type IIA: has $n = 1, 2, 3, 5, 6, 7$ Gauge fields

NS ⊗ NS: $A^{(2)}, A^{(6)}$

R ⊗ R: $A^{(1)}, A^{(3)}, A^{(5)}, A^{(7)}$

→ splits into even and odd

* Note: $\sigma \Rightarrow d\sigma = C^{(1)}$

$$* C^{(1)} = G^{(9)} = d B^{(8)}$$

$$\tilde{e}_{10} = e^{\frac{2}{3}\langle\sigma\rangle}$$

$$(R_{11})_{\text{phys}} = R e^{\frac{1}{3}\langle\sigma\rangle} g_{1010} = \frac{4}{e^{\frac{4}{3}\langle\sigma\rangle}}$$

$$\Rightarrow \left(\frac{(R_{11})_{\text{phys}}}{R} \right)^{3/2} = g_s = e^{\langle\sigma\rangle}$$

• g_s : String coupling of type IIA IOD theory
 $\Rightarrow \alpha_s$

$$R_{11} \rightarrow \infty$$

$$g_s \rightarrow \infty$$

* In the strong coupling limit of type IIA the radius of the circle of comp $\rightarrow \infty$: implying the theory decompactifies.

$$\rightarrow \text{Masses: } \sum_{n=1}^{\infty} \frac{1}{n} \alpha'^2 = \int e^{-2\phi/3} \alpha' \alpha'^2$$

ex: $\square_{11}^2 \phi = 0$

$$= \left(\square_{10}^2 + e^{2\sigma/3} (\partial_{11})^2 \right) \phi$$

$$= \left(\square_{10}^2 + \left(\frac{e^{2\sigma/3}}{R} \right)^2 \right) \phi = 0$$

$$M_n = \frac{e^{2\sigma/3}}{R}$$

$$M_n = \frac{n}{R_{11}}$$

$$e^{2\sigma/3} = (R_{11})/R$$

and so

In the limit $R_{11} \rightarrow \infty$, all the modes are massless

* Note, A_μ couples s.t.

$$(\tilde{e})^\mu_m \partial_\mu \bullet = \tilde{e}^\mu_m \partial_{11} \bullet + \tilde{e}^\mu_m \partial_\mu \bullet$$

$$= -e^{\sigma/12} A_\mu \partial_{11} \bullet + e^{\sigma/12} e^\mu_m \partial_\mu \bullet$$

$$= e^{\sigma/12} e^\mu_m (\partial_\mu - A_\mu \partial_{11}) \bullet$$

$$e^{\sigma/12} e^\mu_m (\partial_\mu - A_\mu \frac{n}{R})$$

Charge: $\frac{n}{R} = q_n$

From Type IIA: $A_\mu \in R \otimes R$ couples with these charges.

→ Solitonic particles with above charges

* Strong Coupling limit of type IIA string theory is an 11D theory called M-theory whose low energy limit is 11D sugra.